AP PRECALCULUS

Unit 1 • Lesson 1.11 practice • Answer key

# Lesson 1.11 • Equivalent Representations of Polynomial and Rational Expressions

5 multiple-choice questions and 1 free-response question • Answer key with explanations

**Directions**

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

### Question 1

NO CALCULATOR

Which is the standard form of $(x+1)^3$?

(A) $x^3+1$

(B) $x^3+3x^2+3x+1$

(C) $x^3+3x+1$

(D) $x^3+2x^2+2x+1$

**Solution**

**Answer: (B).** Row 3 of Pascal's triangle: $1, 3, 3, 1$.

### Question 2

NO CALCULATOR

What is the remainder when $x^3-2x+5$ is divided by $x-2$?

(A) 1

(B) 5

(C) 9

(D) 13

**Solution**

**Answer: (C).** The remainder equals $p(2) = 8-4+5 = 9$.

### Question 3

NO CALCULATOR

Which is the completely factored form of $x^3-9x$?

(A) $x(x-3)(x+3)$

(B) $x(x-3)^2$

(C) $(x-3)(x+3)$

(D) $x(x^2+9)$

**Solution**

**Answer: (A).** $x^3-9x = x(x^2-9) = x(x-3)(x+3)$.

### Question 4

NO CALCULATOR

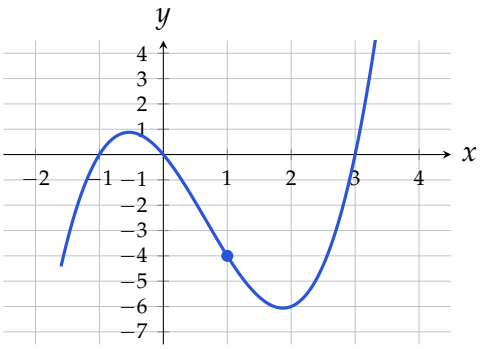
What is the coefficient of $x^3$ in the expansion of $(x+2)^5$?

(A) 10
(B) 20
(C) 80
(D) 40

**Solution**
**Answer: (D).** Row 5 is $1, 5, 10, 10, 5, 1$. The $x^3$ term is $10\,x^3 \cdot 2^2 = 40x^3$.

**Question 5** (graph) NO CALCULATOR

The graph of a cubic polynomial $p$ with leading coefficient 1 is shown. Its zeros are $-1$, 0 and 3. Which is $p(x)$ in standard form?




(A) $x^3 - 2x^2 - 3x$
(B) $x^3 + 2x^2 - 3x$
(C) $-x^3 + 2x^2 + 3x$
(D) $x^3 - 3x$

**Solution**
**Answer: (A).** $p(x) = x(x+1)(x-3) = x(x^2 - 2x - 3) = x^3 - 2x^2 - 3x$. Check with the graph: $p(1) = 1 - 2 - 3 = -4$.

**Question 6** (Free response) NO CALCULATOR

Let $p(x) = 2x^3 + x^2 - 13x + 6$.

(a) Show that $x = 2$ is a zero of $p$.
(b) Write $p(x)$ as a product of linear factors and find all zeros of $p$.
(c) Write $\dfrac{p(x)}{x+1}$ in the form $q(x) + \dfrac{k}{x+1}$.

**Solution**
(a) $p(2) = 16 + 4 - 26 + 6 = 0$.
(b) Synthetic division by 2 on $2, 1, -13, 6$ gives $2, 5, -3$ with remainder 0. So $p(x) = (x-2)(2x^2 + 5x - 3) = (x-2)(2x-1)(x+3)$. Zeros: $2$, $\frac{1}{2}$, $-3$.
(c) Synthetic division by $-1$ gives $2, -1, -12$ with remainder 18: $\dfrac{p(x)}{x+1} = 2x^2 - x - 12 + \dfrac{18}{x+1}$.